

Maxwell Equations

Jake W. Liu

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Chapter Overview

This chapter lays the groundwork for understanding electromagnetic phenomena by introducing Maxwell equations in both integral and differential formulations, highlighting their inherent symmetry and duality. We then explore constitutive relations and matching conditions, crucial for describing field behavior at interfaces. From Maxwell equations, we derive the wave equations and introduce time-harmonic fields for simplified analysis. The chapter concludes with fundamental theorems: Poynting theorem for energy flow, the uniqueness theorem for solution conditions, the reciprocity theorem for source-field interaction symmetry, and the equivalence principle for simplifying radiation and scattering problems.

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Integral and Differential Formulations

At the core of electromagnetic theory lie Maxwell equations and the Lorentz force law. Together with constitutive relations, sources, initial data, and boundary conditions, they form the classical electromagnetic model used in this text. We begin with the *integral formulation* of Faraday's law and Ampère's law, which relates field circulations to fluxes through finite surfaces:

$$\oint_C \vec{E} \cdot d\vec{l} = - \int_S (\partial_t \vec{B} + \vec{M}) \cdot d\vec{s}, \quad (1.1.1)$$

$$\oint_C \vec{H} \cdot d\vec{l} = \int_S (\partial_t \vec{D} + \vec{J}) \cdot d\vec{s}, \quad (1.1.2)$$

Here S is an oriented open surface, $C = \partial S$, and the direction of C follows the right-hand rule relative to $d\vec{s}$. These integral forms represent global field behavior: Faraday's law relates electromotive force to changing magnetic flux and magnetic current, while Ampère's law relates magnetomotive force to changing electric flux and total electric current.

Integral and Differential Formulations (cont.)

These primary equations are supplemented by the continuity equations, which express the conservation of electric and magnetic charge:

$$\oint_S \vec{J} \cdot d\vec{s} = -\partial_t \int_V \rho \, dv, \quad (1.1.3)$$

$$\oint_S \vec{M} \cdot d\vec{s} = -\partial_t \int_V \varrho \, dv, \quad (1.1.4)$$

where these occurrences of $S = \partial V$ are closed surfaces and $d\vec{s}$ points outward, and the Lorentz force equation, which expresses the action of the electromagnetic field on a moving charge and connects the field quantities with measurable mechanical effects:

$$\vec{F} = q (\vec{E} + \vec{v} \times \vec{B}). \quad (1.1.5)$$

The physical quantities in these equations are defined as follows:

- $\vec{E}$: Electric Field Intensity (V/m)
- $\vec{H}$: Magnetic Field Intensity (A/m)
- $\vec{D}$: Electric Flux Density (C/m²)

Integral and Differential Formulations (cont.)

- $\vec{B}$: Magnetic Flux Density (T)
- $\vec{J}$: Volumetric Electric Current Density (A/m²)
- ρ : Electric Charge Density (C/m³)
- $\vec{M}$: Volumetric Magnetic Current Density (V/m²)
- ϱ : Magnetic Charge Density (Wb/m³)

The magnetic sources $\vec{M}$ and ϱ make the equations dual symmetric. No fundamental magnetic charge or associated magnetic current has been observed; later, $\vec{M}$ is also used deliberately as an equivalent mathematical source.

For fields that are differentiable in an open region, the integral surfaces and volumes may be shrunk to obtain the local *differential formulation*. At idealized sheets and material jumps the differential equations are understood distributionally, or as one-sided equations supplemented by the matching conditions derived below.

From (1.1.1)–(1.1.2), applying Stokes' theorem to a piecewise-smooth orientable surface and a sufficiently smooth field,

$$\int_S (\vec{\nabla} \times \vec{\Psi}) \cdot d\vec{s} = \oint_C \vec{\Psi} \cdot d\vec{l}, \quad (1.1.6)$$

Integral and Differential Formulations (cont.)

we obtain the curl-form differential equations:

$$\vec{\nabla} \times \vec{E} = -\partial_t \vec{B} - \vec{M}, \quad (1.1.7)$$

$$\vec{\nabla} \times \vec{H} = \partial_t \vec{D} + \vec{J}. \quad (1.1.8)$$

These represent the local versions of Faraday's and Ampère's laws, expressing how the fields curl in response to time-varying fluxes and sources.

From (1.1.3)–(1.1.4), applying the divergence theorem to a bounded volume with a piecewise-smooth boundary and a sufficiently smooth field,

$$\int_V \vec{\nabla} \cdot \vec{\Psi} dv = \oint_S \vec{\Psi} \cdot d\vec{S}, \quad (1.1.9)$$

we get the continuity equations in differential formulation

$$\vec{\nabla} \cdot \vec{J} = -\partial_t \rho, \quad (1.1.10)$$

$$\vec{\nabla} \cdot \vec{M} = -\partial_t \varrho. \quad (1.1.11)$$

Integral and Differential Formulations

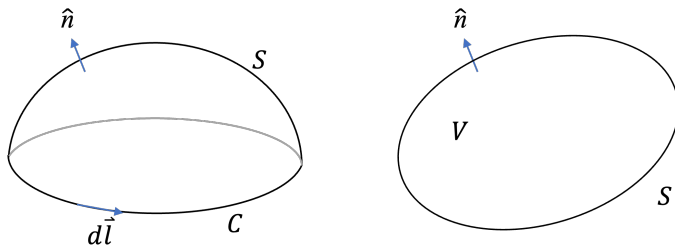


Figure 1.1: Surface orientations: on the open surface, the displayed $d\vec{l}$ direction and $\hat{n}$ obey the right-hand rule; on the closed surface, $\hat{n}$ points outward.

Integral and Differential Formulations

To obtain the divergence forms of Maxwell equations (Gauss' laws), we begin with Ampère's law (1.1.8). Taking the divergence of both sides and using the vector identity $\vec{\nabla} \cdot (\vec{\nabla} \times \vec{H}) \equiv 0$ yields

$$\vec{\nabla} \cdot (\vec{\nabla} \times \vec{H}) = \partial_t (\vec{\nabla} \cdot \vec{D}) + \vec{\nabla} \cdot \vec{J} \equiv 0. \quad (1.1.12)$$

By substituting the continuity relation (1.1.10), we obtain

$$\partial_t (\vec{\nabla} \cdot \vec{D} - \rho) \equiv 0, \quad (1.1.13)$$

which implies that the quantity $(\vec{\nabla} \cdot \vec{D} - \rho)$ remains constant in time:

$$\vec{\nabla} \cdot \vec{D} - \rho = C_D(\vec{r}). \quad (1.1.14)$$

A similar argument applied to Faraday's law leads to

$$\vec{\nabla} \cdot \vec{B} - \varrho = C_B(\vec{r}). \quad (1.1.15)$$

Integral and Differential Formulations (cont.)

The curl and continuity equations determine only that these two constraint residuals are propagated unchanged; they do not determine their initial values. If Gauss' laws hold at one reference time t_0 , then

$$(\vec{\nabla} \cdot \vec{D} - \rho)|_{t=t_0} = 0, \quad (\vec{\nabla} \cdot \vec{B} - \varrho)|_{t=t_0} = 0,$$

so (1.1.13) and its magnetic counterpart imply $C_D(\vec{r}) = C_B(\vec{r}) = 0$ for all time. Gauss' laws are therefore the initial constraints

$$\vec{\nabla} \cdot \vec{D} = \rho, \tag{1.1.16}$$

$$\vec{\nabla} \cdot \vec{B} = \varrho, \tag{1.1.17}$$

and their corresponding integral formulation:

$$\oint_S \vec{D} \cdot d\vec{s} = \int_V \rho \, dv, \tag{1.1.18}$$

$$\oint_S \vec{B} \cdot d\vec{s} = \int_V \varrho \, dv. \tag{1.1.19}$$

Integral and Differential Formulations (cont.)

Thus, Gauss' laws are not consequences of the curl and continuity equations alone: they must be imposed at one time. Once imposed, however, the curl and continuity equations preserve them, so they need not be reimposed during the subsequent time evolution.

Another useful theorem for the curl operator, also derivable from the divergence theorem, is

$$\int_V \vec{\nabla} \times \vec{\Psi} dv = \oint_S \hat{n} \times \vec{\Psi} ds. \quad (1.1.20)$$

Applying this theorem to the curl equations (1.1.7)–(1.1.8) produces

$$\oint_S \hat{n} \times \vec{E} ds = - \int_V (\partial_t \vec{B} + \vec{M}) dv, \quad (1.1.21)$$

$$\oint_S \hat{n} \times \vec{H} ds = \int_V (\partial_t \vec{D} + \vec{J}) dv. \quad (1.1.22)$$

(1.1.21) and (1.1.22) are particularly useful for deriving the matching conditions, as shown in Section 1.2.

Constitutive Relations

For Maxwell equations to form a complete and solvable system, the field vectors must be related through constitutive relations, which describe how the medium responds to electric and magnetic fields. These relations are essential for characterizing the macroscopic electromagnetic properties of materials.

In the absence of the fictitious quantities $\vec{M}$ and ϱ , the system contains five vector quantities— $\vec{E}$, $\vec{H}$, $\vec{D}$, $\vec{B}$, $\vec{J}$ —and one scalar quantity ρ , for 16 scalar field components. The six curl-equation components govern the evolution, while Gauss' laws are the two constraints discussed above and charge continuity is the associated compatibility condition. None of these equations specifies the material response $\vec{D}(\vec{E}, \vec{H})$, $\vec{B}(\vec{E}, \vec{H})$, or the induced part of $\vec{J}(\vec{E}, \vec{H})$. Three vector constitutive relations—nine scalar component relations—supply that missing material information. Schematically, they may be written as

$$\vec{D} = \vec{\bar{C}}_1(\vec{E}, \partial_t \vec{E}, \partial_t^2 \vec{E}, \dots; \vec{H}, \partial_t \vec{H}, \partial_t^2 \vec{H}, \dots), \quad (1.1.23a)$$

$$\vec{B} = \vec{\bar{C}}_2(\vec{E}, \partial_t \vec{E}, \partial_t^2 \vec{E}, \dots; \vec{H}, \partial_t \vec{H}, \partial_t^2 \vec{H}, \dots), \quad (1.1.23b)$$

$$\vec{J} = \vec{\bar{C}}_3(\vec{E}, \partial_t \vec{E}, \partial_t^2 \vec{E}, \dots; \vec{H}, \partial_t \vec{H}, \partial_t^2 \vec{H}, \dots). \quad (1.1.23c)$$

Constitutive Relations (cont.)

Within the spatially local macroscopic model used here, the symbols $\bar{\bar{C}}_j$ in the preceding display denote causal constitutive operators, not merely constant tensors. For example, a linear temporally dispersive response can involve a time convolution; the displayed time derivatives are only a schematic expansion of such a response. Media with spatial dispersion require additional dependence on fields at neighboring points (or, equivalently, on spatial wave vector in a homogeneous medium) and are outside this local model. The following restrictions are logically distinct:

- 1 *Stationary (time invariant)*: the response is unchanged by a common shift of the excitation and observation times; its coefficients have no explicit dependence on the absolute time.
- 2 *Uncoupled electric and magnetic response*: $\bar{\bar{C}}_1$ and $\bar{\bar{C}}_3$ depend only on $\vec{E}$, and $\bar{\bar{C}}_2$ depends only on $\vec{H}$. This excludes local magnetoelectric (bianisotropic) coupling, such as Pasteur-type chiral coupling.
- 3 *Linear*: the response obeys superposition. Linearity excludes products and higher powers of the fields, but it does not by itself exclude time derivatives or dispersive memory.

Constitutive Relations (cont.)

- ④ *Instantaneous (nondispersive)*: the response at time t depends only on the fields at that same time, so no time convolution or time derivatives are present.
- ⑤ *Isotropic*: The tensors $\bar{\bar{C}}_j$ reduce to scalar multiples of the identity tensor, denoted C_j (otherwise, the medium is said to be *anisotropic*).
- ⑥ *Homogeneous*: The coefficients $\bar{\bar{C}}_j$ are spatially uniform.

Unless a section states otherwise, the electric and magnetic responses are assumed stationary and uncoupled. Here a *simple medium* means a linear, instantaneous, isotropic, and homogeneous medium. Later sections explicitly relax homogeneity or allow complex, frequency-dependent parameters when those effects are under study.

For a linear, instantaneous dielectric medium, the constitutive relation between $\vec{D}$ and $\vec{E}$ is

$$\vec{D} = \epsilon_0 \vec{E} + \vec{\mathcal{P}}, \quad (1.1.24)$$

where ϵ_0 is the vacuum permittivity ($\epsilon_0 \approx 8.854\,187\,8188 \times 10^{-12}$ F/m), and

$$\vec{\mathcal{P}} = \epsilon_0 \bar{\bar{\chi}}_e \cdot \vec{E} \quad (1.1.25)$$

is the electric polarization, with $\bar{\bar{\chi}}_e$ denoting the electric susceptibility tensor.

Constitutive Relations (cont.)

An applied field can polarize a dielectric through electronic-cloud distortion, relative ionic displacement, or reorientation of permanent dipoles, depending on the material and frequency. The net electric dipole moment per unit volume is $\vec{P}$. In an isotropic linear medium it is parallel to $\vec{E}$; in an anisotropic medium its direction is set by $\bar{\chi}_e \cdot \vec{E}$. More generally, combining (1.1.24) and (1.1.25) defines

$$\vec{D} = \bar{\epsilon} \cdot \vec{E}, \quad (1.1.26)$$

with

$$\bar{\epsilon} = \epsilon_0 \left(\bar{I} + \bar{\chi}_e \right), \quad (1.1.27)$$

where $\bar{\epsilon}$ is the permittivity tensor and $\bar{I}$ is the identity tensor. In an isotropic material, $\bar{\chi}_e = \chi_e \bar{I}$ and hence $\bar{\epsilon} = \epsilon \bar{I}$, so $\vec{D}$ is parallel to $\vec{E}$.

Similarly, the magnetic flux density relates to the magnetic field as

$$\vec{B} = \mu_0 \vec{H} + \vec{M}, \quad (1.1.28)$$

where μ_0 is the vacuum permeability ($\mu_0 \approx 1.256\,637\,06127 \times 10^{-6}$ H/m, very nearly $4\pi \times 10^{-7}$ H/m in the present SI). The magnetic polarization $\vec{M}$ used here has units of

Constitutive Relations (cont.)

tesla; it is μ_0 times the conventional magnetization (whose units are A/m), and is given by

$$\vec{\mathcal{M}} = \mu_0 \bar{\bar{\chi}}_m \cdot \vec{H}, \quad (1.1.29)$$

with $\bar{\bar{\chi}}_m$ being the magnetic susceptibility tensor. This leads to the compact expression:

$$\vec{B} = \bar{\bar{\mu}} \cdot \vec{H}, \quad (1.1.30)$$

where

$$\bar{\bar{\mu}} = \mu_0 \left(\bar{\bar{I}} + \bar{\bar{\chi}}_m \right), \quad (1.1.31)$$

and $\bar{\bar{\mu}}$ is the permeability tensor.

Finally, the induced conduction-current density in a linear local medium is related to the electric field by Ohm's law:

$$\vec{J} = \bar{\bar{\sigma}} \cdot \vec{E}, \quad (1.1.32)$$

where $\bar{\bar{\sigma}}$ is the conductivity tensor. Equation (1.1.32) describes the induced current; a prescribed impressed current $\vec{J}_i$ must be added separately. In ideal classical vacuum, the conductivity is zero.

Constitutive Relations (cont.)

For a simple medium, the tensors $\bar{\bar{\epsilon}}, \bar{\bar{\mu}}, \bar{\bar{\sigma}}$ reduce to scalar constants ϵ, μ, σ , respectively. In the absence of impressed electric and magnetic sources, the primary curl equations then reduce to

$$\vec{\nabla} \times \vec{E} = -\mu \partial_t \vec{H}, \quad (1.1.33)$$

$$\vec{\nabla} \times \vec{H} = (\sigma + \epsilon \partial_t) \vec{E}. \quad (1.1.34)$$

These constitutive relations serve as essential closures to Maxwell equations, linking material response to field behavior.

Symmetry and Duality

An important feature of Maxwell equations is their dual symmetry, which becomes explicit after introducing the fictitious magnetic charge density ϱ and magnetic current density $\vec{M}$. In a linear medium the total field can be decomposed by superposition into an electric-source field and a magnetic-source field, governed by the following two symmetric sets.

In a simple medium, these two sets of Maxwell equations are as follows:

$$\left\{ \begin{array}{l} \vec{\nabla} \times \vec{E} = -\mu \partial_t \vec{H} \\ \vec{\nabla} \times \vec{H} = \epsilon \partial_t \vec{E} + \vec{J} \\ \vec{\nabla} \cdot (\epsilon \vec{E}) = \rho \\ \vec{\nabla} \cdot (\mu \vec{H}) = 0 \end{array} \right., \quad (\text{electric sources}), \quad (1.1.35a)$$

$$\left\{ \begin{array}{l} \vec{\nabla} \times \vec{E} = -\mu \partial_t \vec{H} - \vec{M} \\ \vec{\nabla} \times \vec{H} = \epsilon \partial_t \vec{E} \\ \vec{\nabla} \cdot (\epsilon \vec{E}) = 0 \\ \vec{\nabla} \cdot (\mu \vec{H}) = \varrho \end{array} \right., \quad (\text{magnetic sources}). \quad (1.1.35b)$$

Symmetry and Duality (cont.)

As a formal relabeling (the primed quantities carry the corresponding dual units), the following transformation interchanges (1.1.35a) and (1.1.35b):

$$\begin{aligned}\vec{E} &\rightarrow \vec{H}', & \vec{H} &\rightarrow -\vec{E}', \\ \vec{J} &\rightarrow \vec{M}', & \vec{M} &\rightarrow -\vec{J}', \\ \rho &\rightarrow \varrho', & \varrho &\rightarrow -\rho', \\ \epsilon &\rightarrow \mu', & \mu &\rightarrow \epsilon'.\end{aligned}\tag{1.1.36}$$

For example, its two curl equations transform step by step as

$$\begin{aligned}\vec{\nabla} \times \vec{E} &= -\mu \partial_t \vec{H} \longrightarrow \vec{\nabla} \times \vec{H}' = \epsilon' \partial_t \vec{E}', \\ \vec{\nabla} \times \vec{H} &= \epsilon \partial_t \vec{E} + \vec{J} \longrightarrow -\vec{\nabla} \times \vec{E}' = \mu' \partial_t \vec{H}' + \vec{M}',\end{aligned}$$

which are precisely the magnetic-source curl equations after multiplying the second line by -1 . For nonzero ϵ and μ , the transformation can instead be made within the same medium by inserting its intrinsic impedance $\eta = \sqrt{\mu/\epsilon}$:

$$\begin{aligned}\vec{E} &\rightarrow \eta \vec{H}', & \vec{H} &\rightarrow -\vec{E}'/\eta \\ \vec{J} &\rightarrow \vec{M}'/\eta, & \vec{M} &\rightarrow -\eta \vec{J}' \\ \rho &\rightarrow \varrho'/\eta, & \varrho &\rightarrow -\eta \rho'.\end{aligned}\tag{1.1.37}$$

Symmetry and Duality (cont.)

Here the impedance factors restore the physical units. The equality $\eta^2 = \mu/\epsilon$ verifies, for instance,

$$\begin{aligned}\vec{\nabla} \times (\eta \vec{H}') &= -\mu \partial_t (-\vec{E}'/\eta), \\ \vec{\nabla} \times \vec{H}' &= \frac{\mu}{\eta^2} \partial_t \vec{E}' \\ &= \epsilon \partial_t \vec{E}'.\end{aligned}$$

This symmetry, known as the *duality principle*, highlights the structural equivalence between electric and magnetic fields in Maxwell equations when magnetic sources are introduced.

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Matching Conditions at Material Interfaces

To derive the matching conditions that govern the behavior of electromagnetic fields at an interface between two different media, we employ the integral forms of Maxwell equations. By considering an infinitesimally small, pillbox-shaped Gaussian surface that straddles the interface shown in Fig. 1.2, we can deduce the constraints on the field components.

Applying (1.1.21) to the pillbox, we obtain

$$\begin{aligned} \hat{n} \times (\vec{E}_1 - \vec{E}_2) \Delta S + \int_{\text{side}} (\hat{t} \times \vec{E}) \, ds \\ = -\Delta S \int_{-h/2}^{h/2} (\partial_t \vec{B} + \vec{M}) \, d\zeta, \end{aligned} \tag{1.2.1}$$

where $\hat{n}$ points from medium 2 to medium 1 and $\hat{t}$ is the outward unit normal on the lateral surface. The fields in this local formula are evaluated at the interface point; the pillbox cross-section is also shrunk so that lateral variation contributes only higher-order terms. To see how the surface integral in (1.1.21) produces this expression, decompose the closed pillbox surface into its three pieces,

$$S = S_{\text{top}} \cup S_{\text{bot}} \cup S_{\text{side}},$$

Matching Conditions at Material Interfaces (cont.)

each carrying the outward-pointing normal of the enclosed volume. The top face lies in medium 1 with outward normal $+\hat{n}$, while the bottom face lies in medium 2 with outward normal $-\hat{n}$; letting ζ measure displacement along $\hat{n}$ from the interface, the two flat faces give

$$\begin{aligned}\oint_{S_{\text{top}} \cup S_{\text{bot}}} \hat{n}' \times \vec{E} \, ds &= (+\hat{n}) \times \vec{E}_1 \Delta S + (-\hat{n}) \times \vec{E}_2 \Delta S \\ &= \hat{n} \times (\vec{E}_1 - \vec{E}_2) \Delta S,\end{aligned}$$

where $\hat{n}'$ is the local outward normal and $\vec{E}_{1,2}$ are the one-sided limiting fields. The remaining lateral band has area Ph , where P is the perimeter of the cross-section ΔS , so its bounded-field contribution is $O(Ph) \rightarrow 0$ as $h \rightarrow 0$. More explicitly, the volume integral is

$$\begin{aligned}\int_V (\partial_t \vec{B} + \vec{M}) \, dv &= \int_{\Delta S} ds \int_{-h/2}^{h/2} (\partial_t \vec{B} + \vec{M}) \, d\zeta \\ &= \Delta S \int_{-h/2}^{h/2} (\partial_t \vec{B} + \vec{M}) \Big|_{\vec{r}_0 + \zeta \hat{n}} d\zeta + o(\Delta S)\end{aligned}$$

Matching Conditions at Material Interfaces (cont.)

as the cross-section shrinks about $\vec{r}_0$. Taking the limit $h \rightarrow 0$, we assume that $\vec{B}$ and $\partial_t \vec{B}$ remain finite within the pillbox, so the $\partial_t \vec{B}$ term is $O(h) \rightarrow 0$, while the tangential magnetic current density $\vec{M}$ may contain a sheet contribution whose integral across $[-h/2, h/2]$ stays finite. In distributional notation this is $\vec{M} = \vec{M}_s \delta(\zeta) + \vec{M}_{\text{reg}}$, where $\vec{M}_{\text{reg}}$ is the regular (bounded) remainder and $\hat{n} \cdot \vec{M}_s = 0$. Dividing by ΔS retains only the sheet current; the surface magnetic current density (units: V/m) is therefore

$$\vec{M}_s = \lim_{h \rightarrow 0} \int_{-h/2}^{h/2} \vec{M} d\zeta, \quad (1.2.2)$$

which leads to the boundary condition for the tangential component of the electric field:

$$\hat{n} \times (\vec{E}_1 - \vec{E}_2) = -\vec{M}_s. \quad (1.2.3)$$

Matching Conditions at Material Interfaces

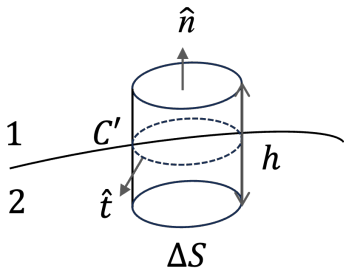


Figure 1.2: The pillbox convention used here: region 1 is above the interface, region 2 is below it, $\hat{n}$ points from 2 to 1, and $\hat{t}$ is outward on the lateral wall.

Matching Conditions at Material Interfaces

Similarly, applying (1.1.22) and taking the limit $h \rightarrow 0$, under the same regularity assumptions, defines the tangential surface electric current density (units: A/m), with $\hat{n} \cdot \vec{J}_s = 0$, as

$$\vec{J}_s = \lim_{h \rightarrow 0} \int_{-h/2}^{h/2} \vec{J} d\zeta, \quad (1.2.4)$$

yielding the boundary condition for the tangential component of the magnetic field:

$$\hat{n} \times (\vec{H}_1 - \vec{H}_2) = \vec{J}_s. \quad (1.2.5)$$

The surface current $\vec{J}_s$ represents any current density whose integral through the vanishing pillbox height remains finite. A perfect electric conductor (PEC) surface is an important example, since the conductor can support an induced surface current; an impressed current sheet may also be prescribed at an interface.

Next, applying the Gauss' law (1.1.18) to the same pillbox gives

$$\hat{n} \cdot (\vec{D}_1 - \vec{D}_2) \Delta S + \int_{\text{side}} \hat{t} \cdot \vec{D} ds = \Delta S \int_{-h/2}^{h/2} \rho d\zeta, \quad (1.2.6)$$

Matching Conditions at Material Interfaces (cont.)

Taking the limit $h \rightarrow 0$, we define the surface charge density (units: C/m^2) as

$$\rho_s = \lim_{h \rightarrow 0} \int_{-h/2}^{h/2} \rho \, d\zeta, \quad (1.2.7)$$

which gives the boundary condition for the normal component of the electric flux density:

$$\hat{n} \cdot (\vec{D}_1 - \vec{D}_2) = \rho_s. \quad (1.2.8)$$

Applying (1.1.19) to the same surface yields the analogous condition for the magnetic flux density:

$$\varrho_s = \lim_{h \rightarrow 0} \int_{-h/2}^{h/2} \varrho \, d\zeta, \quad (1.2.9)$$

$$\hat{n} \cdot (\vec{B}_1 - \vec{B}_2) = \varrho_s. \quad (1.2.10)$$

Finally, applying the continuity equation (1.1.3) to the pillbox gives

$$\hat{n} \cdot (\vec{J}_1 - \vec{J}_2) \Delta S + \int_{\text{side}} \hat{t} \cdot \vec{J} \, ds = -\Delta S \int_{-h/2}^{h/2} \partial_t \rho \, d\zeta. \quad (1.2.11)$$

Matching Conditions at Material Interfaces (cont.)

In the limit $h \rightarrow 0$, the side integral becomes

$$\oint_{C'} \left(\lim_{h \rightarrow 0} \int_{-h/2}^{h/2} \vec{J} d\zeta \right) \cdot \hat{t} dl = \oint_{C'} \vec{J}_s \cdot \hat{t} dl = \int_{S'} \vec{\nabla}_s \cdot \vec{J}_s ds. \quad (1.2.12)$$

where $\vec{\nabla}_s \cdot$ denotes the surface divergence operator, defined as the divergence taken in the tangent plane of the surface. Thus, the continuity condition for the electric current density becomes

$$\hat{n} \cdot (\vec{J}_1 - \vec{J}_2) + \vec{\nabla}_s \cdot \vec{J}_s = -\partial_t \rho_s. \quad (1.2.13)$$

Applying the same procedure to (1.1.4), we obtain the corresponding condition for the magnetic current density:

$$\hat{n} \cdot (\vec{M}_1 - \vec{M}_2) + \vec{\nabla}_s \cdot \vec{M}_s = -\partial_t \varrho_s. \quad (1.2.14)$$

These boundary conditions simplify in several important practical scenarios. Taking medium 2 to be a PEC, the electric field vanishes inside the conductor. At an ordinary PEC boundary with no impressed magnetic-current sheet, $\vec{M}_s = 0$; therefore (1.2.3) gives $\hat{n} \times \vec{E}_1 = 0$. This qualification is important because an equivalent magnetic sheet current is deliberately introduced later in the surface equivalence principle. A perfect

Matching Conditions at Material Interfaces (cont.)

magnetic conductor (PMC), again occupying medium 2, is a hypothetical dual material whose magnetic field vanishes inside. With no impressed electric-current sheet, $\vec{J}_s = 0$, so (1.2.5) gives $\hat{n} \times \vec{H}_1 = 0$.

At an interface between two dielectric media, and assuming explicitly that there is no impressed electric or magnetic current sheet and no free electric or magnetic surface charge, equations (1.2.3)–(1.2.10) then show that the tangential components of $\vec{E}$ and $\vec{H}$, and the normal components of $\vec{D}$ and $\vec{B}$, are continuous. More specifically, a tangential jump of $\vec{E}$ or $\vec{H}$ would require $\vec{M}_s$ or $\vec{J}_s$, respectively, whereas a normal jump of $\vec{D}$ or $\vec{B}$ would require ρ_s or ϱ_s , respectively.

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The Wave Equation

A fundamental implication of Maxwell equations is the existence of electromagnetic waves. These waves arise naturally from the interplay between the electric and magnetic fields described by Maxwell curl equations. To derive the wave equations, we begin by considering a simple medium characterized by constant, nonzero scalar permittivity ϵ and permeability μ . The divisions by ϵ and μ below are not defined in an exact zero-parameter medium; such a limit must instead be taken from the undivided Maxwell equations.

Starting from Faraday's law (1.1.7), and using that μ is a spatially and temporally constant scalar, we take the curl of both sides:

$$\begin{aligned}\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) &= \vec{\nabla} (\vec{\nabla} \cdot \vec{E}) - \nabla^2 \vec{E} \\ &= -\mu \partial_t (\vec{\nabla} \times \vec{H}) - \vec{\nabla} \times \vec{M} \\ &= -\mu \epsilon \partial_t^2 \vec{E} - \mu \partial_t \vec{J} - \vec{\nabla} \times \vec{M},\end{aligned}\tag{1.3.1}$$

where the last line follows from $\vec{\nabla} \times \vec{H} = \epsilon \partial_t \vec{E} + \vec{J}$. Gauss' law supplies the remaining divergence,

$$\vec{\nabla} \cdot \vec{E} = \frac{1}{\epsilon} \vec{\nabla} \cdot \vec{D} = \frac{\rho}{\epsilon}.$$

The Wave Equation (cont.)

Substitution and rearrangement give

$$\nabla^2 \vec{E} - \mu\epsilon\partial_t^2 \vec{E} = \mu\partial_t \vec{J} + \vec{\nabla} \left(\frac{\rho}{\epsilon} \right) + \vec{\nabla} \times \vec{M}. \quad (1.3.2)$$

If $\vec{J} = \sigma\vec{E} + \vec{J}_i$ in a homogeneous conductor, the induced part is not an independent source. Moving it to the left makes the damping term explicit:

$$\nabla^2 \vec{E} - \mu\epsilon\partial_t^2 \vec{E} - \mu\sigma\partial_t \vec{E} = \mu\partial_t \vec{J}_i + \vec{\nabla} \left(\frac{\rho}{\epsilon} \right) + \vec{\nabla} \times \vec{M}. \quad (1.3.3)$$

Equation (1.3.2) is the general inhomogeneous wave equation for $\vec{E}$; its right-hand side contains the total electric current $\vec{J}$, electric charge ρ , and the fictitious or impressed magnetic current $\vec{M}$. In the conductor form immediately above, only the impressed current $\vec{J}_i$ remains on the right because the induced current $\sigma\vec{E}$ has been moved to the left. The magnetic current $\vec{M}$ is distinct from the material magnetic polarization $\vec{\mathcal{M}}$ introduced in (1.1.28).

The Wave Equation (cont.)

For completeness, the magnetic-field equation follows without skipping the intermediate double-curl step:

$$\begin{aligned}\vec{\nabla} \times (\vec{\nabla} \times \vec{H}) &= \vec{\nabla} (\vec{\nabla} \cdot \vec{H}) - \nabla^2 \vec{H} \\ &= \epsilon \partial_t (\vec{\nabla} \times \vec{E}) + \vec{\nabla} \times \vec{J} \\ &= -\mu \epsilon \partial_t^2 \vec{H} - \epsilon \partial_t \vec{M} + \vec{\nabla} \times \vec{J},\end{aligned}$$

and Gauss' law gives $\vec{\nabla} \cdot \vec{H} = \rho/\mu$. Hence

$$\nabla^2 \vec{H} - \mu \epsilon \partial_t^2 \vec{H} = -\vec{\nabla} \times \vec{J} + \vec{\nabla} \left(\frac{\rho}{\mu} \right) + \epsilon \partial_t \vec{M}. \quad (1.3.4)$$

The wave equations (1.3.2) and (1.3.4) can be expressed more compactly using the d'Alembert operator $\square$. Define $v = 1/\sqrt{\mu\epsilon}$, the wave speed (and the phase velocity for a lossless nondispersive medium with real positive μ and ϵ), and then set

$$\square \equiv \nabla^2 - \frac{1}{v^2} \partial_t^2.$$

The Wave Equation (cont.)

Using this notation, the wave equations can be written in a unified matrix form:

$$\square \begin{pmatrix} \vec{E} \\ \vec{H} \end{pmatrix} = \begin{pmatrix} \mu \partial_t \vec{J} + \vec{\nabla} \left(\frac{\rho}{\epsilon} \right) + \vec{\nabla} \times \vec{M} \\ -\vec{\nabla} \times \vec{J} + \vec{\nabla} \left(\frac{\rho}{\mu} \right) + \epsilon \partial_t \vec{M} \end{pmatrix}. \quad (1.3.5)$$

In regions where all source terms vanish, i.e., $\vec{J} = 0$, $\rho = 0$, $\vec{M} = 0$, and $\rho = 0$, the wave equations reduce to their homogeneous forms:

$$\square \vec{E} = 0, \quad \square \vec{H} = 0. \quad (1.3.6)$$

These homogeneous wave equations describe the free propagation of electromagnetic waves in a source-free medium.

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Time-Harmonic Fields and Phasor Notation

In many practical electromagnetic problems, it is convenient to analyze fields that vary sinusoidally with time. A sufficiently well-behaved signal $f(t)$ (or, more generally, a tempered distribution) can be represented as a superposition of time-harmonic components through the inverse Fourier transform:

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} g(\omega) e^{i\omega t} d\omega \quad (1.4.1)$$

with the corresponding time-harmonic components given by:

$$g(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt. \quad (1.4.2)$$

The Fourier and inverse Fourier transform relationship is denoted as

$$\begin{aligned} g(\omega) &= \mathfrak{F}\{f(t)\}, \\ f(t) &= \mathfrak{F}^{-1}\{g(\omega)\}. \end{aligned} \quad (1.4.3)$$

Time-Harmonic Fields and Phasor Notation (cont.)

For a time-harmonic electric field at a chosen positive angular frequency $\omega = 2\pi f > 0$, we can simplify the analysis by using its phasor form:

$$\vec{E}(\vec{r}, t) = \Re \{ \vec{E}(\vec{r}, \omega) e^{i\omega t} \}, \quad (1.4.4)$$

where $\vec{E}(\vec{r}, \omega)$ is a complex vector that contains information about the amplitude and phase of the field and $\Re\{\cdot\}$ denotes taking the real part ($\Im\{\cdot\}$ denotes taking the imaginary part). Notice that the same notation for the field is used for both the time-domain and frequency-domain fields. The functional dependence will be omitted when no ambiguity arises.

In this phasor representation, the time derivative operator ∂_t is simply replaced by multiplication by $i\omega$. This transforms Maxwell equations into the time-harmonic form:

$$\vec{\nabla} \times \vec{E} = -i\omega \vec{B} - \vec{M}, \quad (1.4.5)$$

$$\vec{\nabla} \times \vec{H} = i\omega \vec{D} + \vec{J}, \quad (1.4.6)$$

$$\vec{\nabla} \cdot \vec{D} = \rho, \quad (1.4.7)$$

$$\vec{\nabla} \cdot \vec{B} = \varrho. \quad (1.4.8)$$

Time-Harmonic Fields and Phasor Notation (cont.)

Using the constitutive relations $\vec{D} = \epsilon \vec{E}$ and $\vec{J} = \sigma \vec{E} + \vec{J}_i$ (where $\vec{J}_i$ represents an impressed or external current source), Ampère's law (1.4.6) can be rewritten as:

$$\vec{\nabla} \times \vec{H} = i\omega \left(\epsilon - i\frac{\sigma}{\omega} \right) \vec{E} + \vec{J}_i. \quad (1.4.9)$$

It is convenient to define a complex permittivity

$$\epsilon_c = \epsilon - i\frac{\sigma}{\omega} = \epsilon' - i\epsilon'', \quad \epsilon', \epsilon'' \in \mathbb{R}, \quad (1.4.10)$$

which incorporates both the dielectric polarization and the conductivity of the medium. Here ϵ is the real (lossless) permittivity; if the medium also exhibits polarization loss, then $\epsilon = \epsilon_R - i\epsilon_I$ (with $\epsilon_I \geq 0$) and the two loss mechanisms combine as $\epsilon_c = \epsilon_R - i(\epsilon_I + \sigma/\omega) \equiv \epsilon' - i\epsilon''$, so that $\epsilon' = \epsilon_R$ and $\epsilon'' = \epsilon_I + \sigma/\omega$. For an ordinary dielectric with $\epsilon' = \epsilon_R > 0$, the loss tangent is defined as the ratio of the loss part to the real part of the complex permittivity:

$$\tan \delta = \frac{\epsilon''}{\epsilon'} = \frac{\epsilon_I}{\epsilon_R} + \frac{\sigma}{\omega\epsilon_R}. \quad (1.4.11)$$

Time-Harmonic Fields and Phasor Notation (cont.)

In this time-harmonic framework, Maxwell equations can be expressed in a highly compact and symmetric form:

$$\vec{\nabla} \times \vec{E} = -i\omega\mu\vec{H} - \vec{M}_i, \quad (1.4.12)$$

$$\vec{\nabla} \times \vec{H} = i\omega\epsilon_c\vec{E} + \vec{J}_i. \quad (1.4.13)$$

For a homogeneous medium, replacing ∂_t by $i\omega$ in the source-free wave operator and incorporating conduction into ϵ_c gives the Helmholtz operator

$$\nabla^2 - \mu\epsilon_c(i\omega)^2 = \nabla^2 + \omega^2\mu\epsilon_c \equiv \nabla^2 + k^2, \quad (1.4.14)$$

where

$$k = \omega\sqrt{\mu\epsilon_c} = k_R + ik_I \quad (1.4.15)$$

is the complex wavenumber. With the adopted $e^{i\omega t}$ convention, an outgoing plane wave in the $+z$ direction has spatial factor

$$e^{-ikz} = e^{-i(k_R + ik_I)z} = e^{k_I z} e^{-ik_R z}.$$

Time-Harmonic Fields and Phasor Notation (cont.)

Spatial attenuation in the $+z$ direction selects the square-root branch $k_I \leq 0$. Define the nonnegative attenuation constant $\alpha = -k_I$ and the signed phase constant $\beta = k_R$ (for the ordinary positive-index branch, $\beta \geq 0$). Then

$$k = \beta - i\alpha, \quad \gamma \equiv \alpha + i\beta = ik, \quad \gamma^2 = -\omega^2 \mu \epsilon_c.$$

In a source-free, homogeneous region, Maxwell equations reduce to wave equations for the electric and magnetic fields. Each Cartesian component satisfies the scalar homogeneous Helmholtz equation:

$$\nabla^2 \psi + k^2 \psi = 0. \quad (1.4.16)$$

This decoupling into scalar equations occurs because, in Cartesian coordinates, the vector Laplacian acting on $\vec{E}$ or $\vec{H}$ reduces to scalar Laplacians on individual components since $\hat{x}, \hat{y}, \hat{z}$ are constant vectors throughout space.

However, in cylindrical or spherical coordinates the situation is more intricate. The unit vectors $\hat{\rho}, \hat{\phi}$ in cylindrical coordinates and $\hat{r}, \hat{\theta}, \hat{\phi}$ in spherical coordinates vary with position, so their directions are not fixed throughout space. As a result, the components of $\vec{E}$ and $\vec{H}$ generally satisfy coupled vector wave equations. Only in certain symmetric and separable configurations can individual field components be shown to satisfy

Time-Harmonic Fields and Phasor Notation (cont.)

decoupled scalar Helmholtz equations. In such cases, scalar field solutions can still be used to construct the full vector field solutions through appropriate coordinate-dependent relations. This will be demonstrated in more detail in Chapter 4.

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Poynting Theorem

The Poynting theorem expresses the conservation of electromagnetic energy. It is derived from Maxwell equations using the vector identity:

$$\vec{\nabla} \cdot (\vec{E} \times \vec{H}) = \vec{H} \cdot (\vec{\nabla} \times \vec{E}) - \vec{E} \cdot (\vec{\nabla} \times \vec{H}). \quad (1.5.1)$$

Substituting Maxwell equations first gives the complete intermediate chain

$$\begin{aligned} \vec{\nabla} \cdot (\vec{E} \times \vec{H}) &= \vec{H} \cdot (-\partial_t \vec{B} - \vec{M}) - \vec{E} \cdot (\partial_t \vec{D} + \vec{J}) \\ &= -\vec{H} \cdot \partial_t \vec{B} - \vec{E} \cdot \partial_t \vec{D} - \vec{H} \cdot \vec{M} - \vec{E} \cdot \vec{J}. \end{aligned}$$

For linear, nondispersive media with time-independent symmetric constitutive tensors,

$$\begin{aligned} \vec{E} \cdot \partial_t \vec{D} &= \vec{E} \cdot \bar{\epsilon} \cdot \partial_t \vec{E} = \partial_t \left(\frac{1}{2} \vec{E} \cdot \vec{D} \right), \\ \vec{H} \cdot \partial_t \vec{B} &= \vec{H} \cdot \bar{\mu} \cdot \partial_t \vec{H} = \partial_t \left(\frac{1}{2} \vec{H} \cdot \vec{B} \right). \end{aligned}$$

The identities in the preceding display require symmetry, but their interpretation as nonnegative stored energies additionally requires $\bar{\epsilon}$ and $\bar{\mu}$ to be positive definite. Under the stated algebraic hypotheses, the differential instantaneous Poynting theorem is therefore

Poynting Theorem (cont.)

$$\vec{\nabla} \cdot \vec{S} = -\partial_t \left(\frac{1}{2} \vec{E} \cdot \vec{D} + \frac{1}{2} \vec{H} \cdot \vec{B} \right) - \vec{H} \cdot \vec{M} - \vec{E} \cdot \vec{J}, \quad (1.5.2)$$

or equivalently,

$$\vec{\nabla} \cdot \vec{S} = -\partial_t (w_e + w_m) - p_l, \quad (1.5.3)$$

where $\vec{S} = \vec{E} \times \vec{H}$ is the instantaneous Poynting vector (W/m^2); $w_e = \frac{1}{2} \vec{E} \cdot \vec{D}$ and $w_m = \frac{1}{2} \vec{H} \cdot \vec{B}$ are the instantaneous electric and magnetic quadratic energy terms (and are nonnegative stored-energy densities when the constitutive tensors are positive definite); and $p_l = \vec{H} \cdot \vec{M} + \vec{E} \cdot \vec{J}$ is the instantaneous rate of energy transfer from the field to the currents per unit volume. It is a loss when positive and a gain when negative. For nonlinear or temporally dispersive media, the first intermediate line remains valid, but the stored energy cannot in general be replaced by the two simple quadratic forms. Integrating this expression over a volume V and applying the divergence theorem yields the integral form:

$$\oint_S \vec{S} \cdot d\vec{s} = -\partial_t \int_V (w_e + w_m) dv - \int_V p_l dv. \quad (1.5.4)$$

Poynting Theorem (cont.)

In the case of time-harmonic fields, it is more practical to work with time-averaged quantities. Assuming fields of the form

$\vec{E}(\vec{r}, t) = \Re\{\vec{E}(\vec{r})e^{i\omega t}\} = [\vec{E}(\vec{r})e^{i\omega t} + \vec{E}^*(\vec{r})e^{-i\omega t}]/2$, the time-dependent Poynting vector becomes:

$$\vec{S}(\vec{r}, t) = \frac{1}{2}\Re\{\vec{E} \times \vec{H}^*\} + \frac{1}{2}\Re\{\vec{E} \times \vec{H}e^{2i\omega t}\}. \quad (1.5.5)$$

Here, the first term represents a steady (time-invariant) component, while the second term oscillates at frequency 2ω . The 2ω term does not contribute to the net energy transfer over a full period and hence averages to zero. Thus, the time-averaged Poynting vector over one period $T = 2\pi/\omega$ is:

$$\begin{aligned} \vec{S}_{\text{avg}} &= \frac{1}{T} \int_0^T \vec{E}(\vec{r}, t) \times \vec{H}(\vec{r}, t) dt \\ &= \frac{1}{2}\Re\{\vec{E} \times \vec{H}^*\} = \Re\{\vec{S}_c\}, \end{aligned} \quad (1.5.6)$$

where $\vec{S}_{\text{avg}}$ is the time-averaged Poynting vector and

Poynting Theorem (cont.)

$$\vec{S}_c = \frac{1}{2} \vec{E} \times \vec{H}^* \quad (1.5.7)$$

is defined as the complex Poynting vector. This leads to the complex Poynting theorem. To obtain it, we repeat the construction of the instantaneous theorem with the conjugated cross product. Applying the vector identity (1.5.1) to $\vec{E} \times \vec{H}^*$,

$$\vec{\nabla} \cdot \vec{S}_c = \frac{1}{2} \vec{\nabla} \cdot (\vec{E} \times \vec{H}^*) = \frac{1}{2} \left[\vec{H}^* \cdot (\vec{\nabla} \times \vec{E}) - \vec{E} \cdot (\vec{\nabla} \times \vec{H}^*) \right].$$

We now substitute Faraday's law (1.4.12) for $\vec{\nabla} \times \vec{E}$ and the conjugate of Ampère's law (1.4.13) for $\vec{\nabla} \times \vec{H}^*$. Since the spatial derivative operator is real, $(\vec{\nabla} \times \vec{H})^* = \vec{\nabla} \times \vec{H}^*$, and hence

$$\vec{\nabla} \times \vec{E} = -i\omega\mu\vec{H} - \vec{M}_i, \quad \vec{\nabla} \times \vec{H}^* = -i\omega\epsilon_c^*\vec{E}^* + \vec{J}_i^*.$$

Inserting these equations and using $\vec{H}^* \cdot \vec{H} = |\vec{H}|^2$ and $\vec{E} \cdot \vec{E}^* = |\vec{E}|^2$ gives the complex Poynting theorem

$$\vec{\nabla} \cdot \vec{S}_c = \frac{i\omega}{2} (\epsilon_c^* |\vec{E}|^2 - \mu |\vec{H}|^2) - \frac{1}{2} (\vec{E} \cdot \vec{J}_i^* + \vec{H}^* \cdot \vec{M}_i).$$

Poynting Theorem (cont.)

To expose its physical content without treating complex quantities as stored energies, write the passive scalar parameters using the $e^{i\omega t}$ convention as

$$\epsilon_c = \epsilon' - i\epsilon'', \quad \mu = \mu' - i\mu'', \quad \epsilon'', \mu'' \geq 0.$$

Define the two real quadratic combinations

$$\begin{aligned} A &= \epsilon' |\vec{E}|^2 - \mu' |\vec{H}|^2, \\ B &= \epsilon'' |\vec{E}|^2 + \mu'' |\vec{H}|^2. \end{aligned} \tag{1.5.8}$$

Then every algebraic step in the material term is

$$\begin{aligned} \frac{i\omega}{2} (\epsilon_c^* |\vec{E}|^2 - \mu |\vec{H}|^2) &= \frac{i\omega}{2} (A + iB) \\ &= -\frac{\omega}{2} B + i\frac{\omega}{2} A. \end{aligned} \tag{1.5.9}$$

Poynting Theorem (cont.)

Consequently,

$$\begin{aligned}\vec{\nabla} \cdot \vec{S}_c &= -p_{\text{mat}} \\ &\quad + i2\omega \left(w_e^{(r)} - w_m^{(r)} \right) \\ &\quad - p_{\text{src,c}}, \\ p_{\text{mat}} &= \frac{\omega}{2} \left(\epsilon'' |\vec{E}|^2 + \mu'' |\vec{H}|^2 \right), \\ w_e^{(r)} &= \frac{\epsilon' |\vec{E}|^2}{4}, \\ w_m^{(r)} &= \frac{\mu' |\vec{H}|^2}{4}, \\ p_{\text{src,c}} &= \frac{1}{2} \vec{E} \cdot \vec{J}_i^* + \frac{1}{2} \vec{H}^* \cdot \vec{M}_i.\end{aligned}\tag{1.5.10}$$

Taking the real part yields the time-averaged power balance

$$\vec{\nabla} \cdot \vec{S}_{\text{avg}} = -p_{\text{mat}} - \Re\{p_{\text{src,c}}\}.\tag{1.5.11}$$

Thus $p_{\text{mat}} \geq 0$ is the material dissipation density. In a lossless, nondispersive scalar medium with $\epsilon' > 0$ and $\mu' > 0$, $w_e^{(r)}$ and $w_m^{(r)}$ are the time-averaged stored electric and

Poynting Theorem (cont.)

magnetic energy densities. In a dispersive medium the complex Poynting identity remains valid, but stored energy requires the appropriate dispersion-dependent expression; it is not generally $\frac{1}{4}\vec{E} \cdot \vec{D}^*$ or $\frac{1}{4}\vec{H} \cdot \vec{B}^*$.

The Uniqueness Theorem

Consider a bounded volume V at a real nonzero frequency ω , with the same scalar (possibly complex) ϵ and μ , the same sources, and the same boundary data. Suppose two solutions $(\vec{E}_1, \vec{H}_1)$ and $(\vec{E}_2, \vec{H}_2)$ exist. Their governing equations are

$$\begin{cases} \vec{\nabla} \times \vec{E}_1 = -i\omega\mu\vec{H}_1 - \vec{M} \\ \vec{\nabla} \times \vec{H}_1 = i\omega\epsilon\vec{E}_1 + \vec{J} \end{cases}, \quad \begin{cases} \vec{\nabla} \times \vec{E}_2 = -i\omega\mu\vec{H}_2 - \vec{M} \\ \vec{\nabla} \times \vec{H}_2 = i\omega\epsilon\vec{E}_2 + \vec{J} \end{cases}. \quad (1.5.12)$$

We then define the difference fields as $\delta\vec{E} = \vec{E}_1 - \vec{E}_2$ and $\delta\vec{H} = \vec{H}_1 - \vec{H}_2$. These difference fields must satisfy the source-free (homogeneous) Maxwell equations:

$$\begin{cases} \vec{\nabla} \times \delta\vec{E} = -i\omega\mu\delta\vec{H} \\ \vec{\nabla} \times \delta\vec{H} = i\omega\epsilon\delta\vec{E} \end{cases}. \quad (1.5.13)$$

Now, let us dot the first equation in (1.5.13) with $\delta\vec{H}^*$ and the conjugate of the second with $\delta\vec{E}$, and subtract the two:

$$\begin{aligned} & \vec{\nabla} \times \delta\vec{E} \cdot \delta\vec{H}^* - \vec{\nabla} \times \delta\vec{H}^* \cdot \delta\vec{E} \\ &= \vec{\nabla} \cdot (\delta\vec{E} \times \delta\vec{H}^*) = -i\omega (\mu|\delta\vec{H}|^2 - \epsilon^*|\delta\vec{E}|^2). \end{aligned} \quad (1.5.14)$$

The Uniqueness Theorem (cont.)

By integrating over the volume V and applying the divergence theorem (1.1.9), we derive the following integral relation:

$$\oint_S (\delta \vec{E} \times \delta \vec{H}^*) \cdot d\vec{s} = -i\omega \int_V (\mu |\delta \vec{H}|^2 - \epsilon^* |\delta \vec{E}|^2) dv. \quad (1.5.15)$$

The key insight is that if the surface integral in (1.5.15) is zero, then the volume integral must also be zero, which implies that:

$$-i\omega \int_V (\mu |\delta \vec{H}|^2 - \epsilon^* |\delta \vec{E}|^2) dv = 0.$$

Substituting $\mu = \Re\{\mu\} + i \Im\{\mu\}$ and $\epsilon^* = \Re\{\epsilon\} - i \Im\{\epsilon\}$, the integrand separates into its real and imaginary parts,

$$\begin{aligned} \mu |\delta \vec{H}|^2 - \epsilon^* |\delta \vec{E}|^2 &= (\Re\{\mu\} |\delta \vec{H}|^2 - \Re\{\epsilon\} |\delta \vec{E}|^2) \\ &\quad + i (\Im\{\mu\} |\delta \vec{H}|^2 + \Im\{\epsilon\} |\delta \vec{E}|^2). \end{aligned}$$

The Uniqueness Theorem (cont.)

The leading factor $-i$ interchanges these two parts, since $-i\omega(A + iB) = \omega B - i\omega A$. Hence the vanishing condition becomes

$$\begin{aligned} & \omega \int_V (\Im\{\mu\}|\delta\vec{H}|^2 + \Im\{\epsilon\}|\delta\vec{E}|^2) dv \\ & - i\omega \int_V (\Re\{\mu\}|\delta\vec{H}|^2 - \Re\{\epsilon\}|\delta\vec{E}|^2) dv = 0. \end{aligned}$$

Because ω is real, the real and imaginary parts must each vanish on their own, which yields

$$\begin{cases} \int_V (\Re\{\mu\}|\delta\vec{H}|^2 - \Re\{\epsilon\}|\delta\vec{E}|^2) dv = 0 \\ \int_V (\Im\{\mu\}|\delta\vec{H}|^2 + \Im\{\epsilon\}|\delta\vec{E}|^2) dv = 0 \end{cases}. \quad (1.5.16)$$

With the $e^{i\omega t}$ convention, a sufficient strict-dissipation condition is $\Im\{\epsilon\} < 0$ and $\Im\{\mu\} < 0$ throughout V . The integrand in the second line of (1.5.16) is then nonpositive and is strictly negative wherever either difference field is nonzero. Its integral can vanish only when $\delta\vec{E} = \delta\vec{H} = 0$ almost everywhere (and, by the assumed

The Uniqueness Theorem (cont.)

field regularity, everywhere), establishing uniqueness. For the surface integral on the left-hand side of (1.5.15) to be zero, vector identities yield the following equation:

$$\begin{aligned}\oint_S (\delta \vec{E} \times \delta \vec{H}^*) \cdot d\vec{s} &= \oint_S \hat{n} \times \delta \vec{E} \cdot \delta \vec{H}^* ds \\ &= \oint_S \delta \vec{H}^* \times \hat{n} \cdot \delta \vec{E} ds = 0.\end{aligned}\tag{1.5.17}$$

This condition is satisfied if any of the following boundary conditions are specified on the surface S enclosing the volume V :

- 1 The tangential component of $\vec{E}$ is specified over the entire surface S (i.e., $\hat{n} \times \delta \vec{E} = 0$).
- 2 The tangential component of $\vec{H}$ is specified over the entire surface S (i.e., $\hat{n} \times \delta \vec{H} = 0$).
- 3 The tangential component of $\vec{E}$ is specified over a portion of S , and the tangential component of $\vec{H}$ is specified over the remainder of the surface.

The Uniqueness Theorem (cont.)

Thus the stated boundary data give uniqueness under strict dissipation. In a closed lossless cavity, the dissipative integral does not prove that the difference fields vanish. At an eigenfrequency a nonzero source-free eigenmode satisfies the homogeneous boundary data and may be added to a solution. Away from the spectrum, uniqueness requires the corresponding lossless boundary-value argument; it is not established by setting a zero dissipative integral equal to zero.

For a connected homogeneous lossless exterior with real positive ϵ, μ , a bounded PEC or PMC obstacle, and the outgoing condition, Section 3.1.3 supplies the missing proof: zero real power implies zero far-field amplitude, and an outgoing homogeneous solution with zero far field is then shown to vanish. The boundary condition at infinity is part of that problem. Other exterior media and boundary conditions require their own hypotheses; limiting absorption is a solution-selection procedure, not a substitute for proving uniqueness in an unspecified problem class. These distinctions apply when using the equivalence principle in subsection 1.5.4 and the image constructions of Chapter 3.

The Reciprocity Theorem

The reciprocity theorem is a powerful principle that relates the fields produced by two different sets of source distributions within the same simple medium. Let us consider two sets of sources, $(\vec{J}_1, \vec{M}_1)$ and $(\vec{J}_2, \vec{M}_2)$, which generate the fields $(\vec{E}_1, \vec{H}_1)$ and $(\vec{E}_2, \vec{H}_2)$, respectively. We can define the *reaction* of field 1 on source 2, and vice versa, as:

$$\langle 1, 2 \rangle = \int_V (\vec{E}_1 \cdot \vec{J}_2 - \vec{H}_1 \cdot \vec{M}_2) dv, \quad (1.5.18a)$$

$$\langle 2, 1 \rangle = \int_V (\vec{E}_2 \cdot \vec{J}_1 - \vec{H}_2 \cdot \vec{M}_1) dv. \quad (1.5.18b)$$

From (1.4.12) and (1.4.13), we have the following set of equations:

$$\vec{\nabla} \times \vec{E}_1 = -i\omega\mu\vec{H}_1 - \vec{M}_1, \quad (1.5.19a)$$

$$\vec{\nabla} \times \vec{H}_1 = i\omega\epsilon_c\vec{E}_1 + \vec{J}_1, \quad (1.5.19b)$$

$$\vec{\nabla} \times \vec{E}_2 = -i\omega\mu\vec{H}_2 - \vec{M}_2, \quad (1.5.19c)$$

$$\vec{\nabla} \times \vec{H}_2 = i\omega\epsilon_c\vec{E}_2 + \vec{J}_2. \quad (1.5.19d)$$

The Reciprocity Theorem (cont.)

By dotting (1.5.19a) with $\vec{H}_2$ and (1.5.19d) with $\vec{E}_1$ and then subtracting the two, we proceed as follows. We invoke the same vector identity used for the Poynting theorem, namely

$$\vec{\nabla} \cdot (\vec{E}_1 \times \vec{H}_2) = \vec{H}_2 \cdot (\vec{\nabla} \times \vec{E}_1) - \vec{E}_1 \cdot (\vec{\nabla} \times \vec{H}_2).$$

Substituting (1.5.19a) for $\vec{\nabla} \times \vec{E}_1$ and (1.5.19d) for $\vec{\nabla} \times \vec{H}_2$,

$$\begin{aligned}\vec{\nabla} \cdot (\vec{E}_1 \times \vec{H}_2) &= \vec{H}_2 \cdot (-i\omega\mu\vec{H}_1 - \vec{M}_1) \\ &\quad - \vec{E}_1 \cdot (i\omega\epsilon_c\vec{E}_2 + \vec{J}_2) \\ &= -i\omega\mu\vec{H}_1 \cdot \vec{H}_2 - \vec{H}_2 \cdot \vec{M}_1 \\ &\quad - i\omega\epsilon_c\vec{E}_1 \cdot \vec{E}_2 - \vec{E}_1 \cdot \vec{J}_2.\end{aligned}$$

Negating both sides then yields:

$$-\vec{\nabla} \cdot (\vec{E}_1 \times \vec{H}_2) = i\omega\mu\vec{H}_1 \cdot \vec{H}_2 + \vec{H}_2 \cdot \vec{M}_1 + i\omega\epsilon_c\vec{E}_1 \cdot \vec{E}_2 + \vec{E}_1 \cdot \vec{J}_2. \quad (1.5.20a)$$

Then, by interchanging the subscripts 1 and 2, we get:

$$-\vec{\nabla} \cdot (\vec{E}_2 \times \vec{H}_1) = i\omega\mu\vec{H}_2 \cdot \vec{H}_1 + \vec{H}_1 \cdot \vec{M}_2 + i\omega\epsilon_c\vec{E}_2 \cdot \vec{E}_1 + \vec{E}_2 \cdot \vec{J}_1. \quad (1.5.20b)$$

The Reciprocity Theorem (cont.)

Subtracting the two equations allows us to derive the Lorentz reciprocity theorem in its differential form:

$$\begin{aligned}\vec{\nabla} \cdot (\vec{E}_2 \times \vec{H}_1 - \vec{E}_1 \times \vec{H}_2) \\ = (\vec{E}_1 \cdot \vec{J}_2 - \vec{H}_1 \cdot \vec{M}_2) - (\vec{E}_2 \cdot \vec{J}_1 - \vec{H}_2 \cdot \vec{M}_1).\end{aligned}\tag{1.5.21}$$

Integrating this equation over a volume V and applying the divergence theorem, we obtain the integral form of the theorem:

$$\oint_S (\vec{E}_2 \times \vec{H}_1 - \vec{E}_1 \times \vec{H}_2) \cdot d\vec{s} = \langle 1, 2 \rangle - \langle 2, 1 \rangle.\tag{1.5.22}$$

If S is taken to infinity through a common homogeneous exterior medium and both fields satisfy the outgoing Sommerfeld radiation condition, the surface integral becomes zero. This is not merely because the fields vanish: radiating fields fall as $1/r$, while the area grows as r^2 . To leading order each outgoing field obeys $\vec{H}_j = \eta^{-1} \hat{r} \times \vec{E}_j + O(r^{-2})$. Therefore

$$\begin{aligned}\hat{r} \cdot (\vec{E}_2 \times \vec{H}_1 - \vec{E}_1 \times \vec{H}_2) &= \frac{1}{\eta} [\vec{E}_2 \cdot \vec{E}_1 - \vec{E}_1 \cdot \vec{E}_2] + O(r^{-3}) \\ &= O(r^{-3}),\end{aligned}$$

The Reciprocity Theorem (cont.)

whose integral over a sphere vanishes as $r \rightarrow \infty$. This leaves the Rayleigh–Carson reciprocity theorem:

$$\langle 1, 2 \rangle = \langle 2, 1 \rangle. \quad (1.5.23)$$

This theorem expresses a fundamental symmetry: the reaction of field 1 on source 2 is identical to the reaction of field 2 on source 1. In a source-free region, we can further simplify (1.5.22) to

$$\oint_S (\vec{E}_2 \times \vec{H}_1 - \vec{E}_1 \times \vec{H}_2) \cdot d\vec{s} = 0. \quad (1.5.24)$$

For example, apply (1.5.24) to a source-free PEC waveguide segment between cross-sections $A(z_1)$ and $A(z_2)$. The lateral-wall integral vanishes because the electric fields are normal to the PEC wall, leaving

$$\begin{aligned} \mathcal{R}(z) &\equiv \int_{A(z)} (\vec{E}_2 \times \vec{H}_1) \cdot \hat{z} dA \\ &\quad - \int_{A(z)} (\vec{E}_1 \times \vec{H}_2) \cdot \hat{z} dA, \\ \mathcal{R}(z_2) &= \mathcal{R}(z_1). \end{aligned}$$

The Reciprocity Theorem (cont.)

Thus this bilinear modal overlap is conserved along a uniform source-free guide; the usual orthogonality relations follow after inserting the distinct modal z dependences. As a corollary of (1.5.22), an impressed tangential electric current colocated with a PEC does not radiate into the exterior; the dual statement holds for a tangential magnetic current on a PMC. Let $\vec{J}_1$ be the PEC-supported sheet distribution and place an arbitrary test current element at an exterior point $\vec{r}$,

$$\vec{J}_2(\vec{r}') = I d\ell \hat{a} \delta(\vec{r}' - \vec{r}).$$

Then reciprocity gives

$$\begin{aligned} \langle 1, 2 \rangle &= \int_V \vec{E}_1 \cdot \vec{J}_2 d\nu = I d\ell \vec{E}_1(\vec{r}) \cdot \hat{a} \\ &= \langle 2, 1 \rangle = \int_V \vec{E}_2 \cdot \vec{J}_1 d\nu = 0. \end{aligned} \tag{1.5.25}$$

The last equality holds because $\vec{E}_2$ generated by $\vec{J}_2$ has zero tangential component on the PEC, while $\vec{J}_1$ is tangential and supported on that surface. Equation (1.5.25) gives $\vec{E}_1(\vec{r}) \cdot \hat{a} = 0$. Since the exterior observation point $\vec{r}$ and test-dipole orientation $\hat{a}$ are

The Reciprocity Theorem (cont.)

arbitrary, $\vec{E}_1 = 0$ throughout the exterior region: the tangential impressed electric current colocated with the PEC does not radiate into that region. The dual argument shows that a tangential impressed magnetic current colocated with a PMC does not radiate into its exterior.

The Reciprocity Theorem

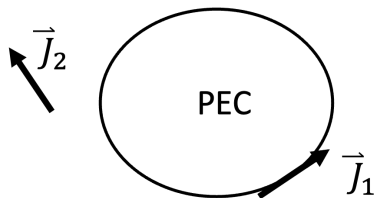


Figure 1.3: A PEC-supported tangential sheet current $\vec{J}_1$ and an arbitrary exterior test-current element $\vec{J}_2$, as used in (1.5.25).

The Reciprocity Theorem

The displayed derivation uses scalar ϵ_c and μ . It extends pointwise to reciprocal inhomogeneous anisotropic media provided the constitutive tensors are transpose-symmetric, $\bar{\bar{\epsilon}}_c^T = \bar{\bar{\epsilon}}_c$ and $\bar{\bar{\mu}}^T = \bar{\bar{\mu}}$. Hermitian symmetry is not required, so reciprocal loss is allowed. For nonsymmetric (nonreciprocal) tensors, the material terms do not cancel and this reciprocity identity does not follow.

Equivalence Principle

The equivalence principle allows the actual sources within a region to be replaced by an equivalent distribution on its boundary without changing the exterior field. Its validity follows from the uniqueness theorem: once an appropriate tangential boundary datum (electric or magnetic), the exterior sources, and the outgoing radiation condition are specified, the exterior solution is unique. Two common forms are the volume and surface equivalence principles.

A. Volume Equivalence Principle. Let the sources $(\vec{J}, \vec{M})$ radiate in free space, whose constitutive parameters are ϵ_0 and μ_0 :

$$\begin{cases} \vec{\nabla} \times \vec{E}_0 = -i\omega\mu_0\vec{H}_0 - \vec{M}, \\ \vec{\nabla} \times \vec{H}_0 = i\omega\epsilon_0\vec{E}_0 + \vec{J}. \end{cases} \quad (1.5.26)$$

Now place a material object with permittivity ϵ and permeability μ in that background; outside the object the parameters retain the background values ϵ_0, μ_0 . The total fields satisfy, pointwise,

$$\begin{cases} \vec{\nabla} \times \vec{E} = -i\omega\mu\vec{H} - \vec{M}, \\ \vec{\nabla} \times \vec{H} = i\omega\epsilon\vec{E} + \vec{J}. \end{cases} \quad (1.5.27)$$

Equivalence Principle (cont.)

Subtracting the vacuum field equations from those with the material yields the equations governing the scattered fields ($\vec{E}_s = \vec{E} - \vec{E}_0$, $\vec{H}_s = \vec{H} - \vec{H}_0$):

$$\begin{cases} \vec{\nabla} \times \vec{E}_s = -i\omega(\mu\vec{H} - \mu_0\vec{H}_0), \\ \vec{\nabla} \times \vec{H}_s = i\omega(\epsilon\vec{E} - \epsilon_0\vec{E}_0). \end{cases} \quad (1.5.28)$$

The coefficient regrouping that produces background-medium equations is

$$\begin{aligned} -i\omega(\mu\vec{H} - \mu_0\vec{H}_0) &= -i\omega\mu_0(\vec{H} - \vec{H}_0) - i\omega(\mu - \mu_0)\vec{H}, \\ i\omega(\epsilon\vec{E} - \epsilon_0\vec{E}_0) &= +i\omega\epsilon_0(\vec{E} - \vec{E}_0) + i\omega(\epsilon - \epsilon_0)\vec{E}. \end{aligned} \quad (1.5.29)$$

Using $\vec{E}_s = \vec{E} - \vec{E}_0$ and $\vec{H}_s = \vec{H} - \vec{H}_0$, these equations become

$$\begin{cases} \vec{\nabla} \times \vec{E}_s = -i\omega\mu_0\vec{H}_s - \vec{M}_{eq}, \\ \vec{\nabla} \times \vec{H}_s = i\omega\epsilon_0\vec{E}_s + \vec{J}_{eq}, \end{cases} \quad (1.5.30)$$

where the equivalent volume sources are defined as:

$$\begin{cases} \vec{M}_{eq} = i\omega (\mu - \mu_0) \vec{H}, \\ \vec{J}_{eq} = i\omega (\epsilon - \epsilon_0) \vec{E}. \end{cases} \quad (1.5.31)$$

Equivalence Principle (cont.)

These equivalent currents are non-zero only within the volume of the inhomogeneity and generate the same scattered fields as those produced by the original material object.

B. Surface Equivalence Principle. The surface equivalence principle states that the electromagnetic effect of sources inside a closed surface can be exactly replicated outside that surface by an appropriate distribution of equivalent surface currents.

Referring to the configuration on the left-hand side of Fig. 1.4, consider a region V_2 containing sources or material inhomogeneities. If interest is restricted to the fields external to V_2 , one may replace all internal sources with surface-equivalent currents defined on the boundary of V_2 . Based on the tangential matching conditions (1.2.3) and (1.2.5), the equivalent surface currents are given by:

$$\begin{cases} \vec{J}_s = \hat{n} \times (\vec{H}_1 - \vec{H}'_2), \\ \vec{M}_s = -\hat{n} \times (\vec{E}_1 - \vec{E}'_2), \end{cases} \quad (1.5.32)$$

where $\hat{n}$ is the outward normal to the surface of V_2 , $(\vec{E}_1, \vec{H}_1)$ are the fields in the original configuration, and $(\vec{E}'_2, \vec{H}'_2)$ are the fields inside V_2 when only the equivalent surface currents are present.

Equivalence Principle

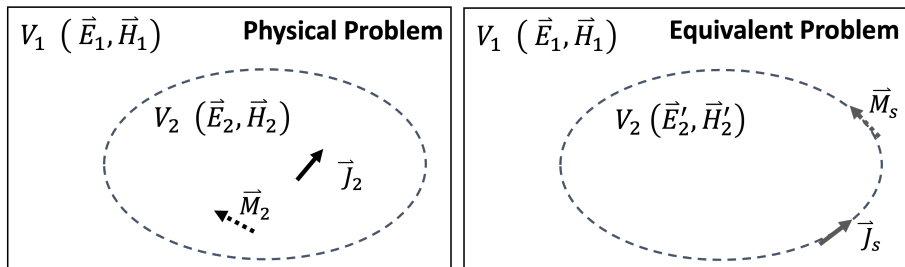


Figure 1.4: Surface equivalence. The normal $\hat{n}$ (not drawn) points outward from V_2 into V_1 ; the solid and dashed boundary arrows denote $\vec{J}_s$ and $\vec{M}_s$, whose signs are fixed by the jump relations in the text.

Equivalence Principle

Three notable special cases are commonly employed in practical applications:

- ① *Love Equivalence Principle*: Setting $\vec{E}'_2 = \vec{H}'_2 = 0$ directly in the two jump relations gives

$$\begin{cases} \vec{J}_s = \hat{n} \times \vec{H}_1, \\ \vec{M}_s = -\hat{n} \times \vec{E}_1. \end{cases} \quad (1.5.33a)$$

- ② *PEC Replacement*: On a PEC, $\hat{n} \times \vec{E}'_2 = 0$, so the magnetic jump relation fixes $\vec{M}_s = -\hat{n} \times \vec{E}_1$. Any accompanying tangential electric current on the PEC produces no exterior field by (1.5.25), and it may therefore be omitted. A sufficient one-current representation is

$$\begin{cases} \vec{J}_s = 0, \\ \vec{M}_s = -\hat{n} \times \vec{E}_1. \end{cases} \quad (1.5.33b)$$

- ③ *PMC Replacement*: Dually, $\hat{n} \times \vec{H}'_2 = 0$, and a tangential magnetic current on the PMC produces no exterior field. A sufficient one-current representation is

$$\begin{cases} \vec{J}_s = \hat{n} \times \vec{H}_1, \\ \vec{M}_s = 0. \end{cases} \quad (1.5.33c)$$

Equivalence Principle (cont.)

In practice the surface formulation is commonly preferred for its reduced dimensionality, which simplifies the treatment of radiation and scattering problems.

Further Reading

For a concise summary of Maxwell equations, see Barkeshli [3]. The treatment of the fundamental theorems follows standard references by Harrington [11], Balanis [2], and Jin [15]. Someda [27] adds an optics perspective, including a focused discussion of polarization.

Although vector algebra is the most familiar framework for Maxwell equations, differential forms provide a compact geometric alternative; see the treatments by Deschamps [8] and Lindell [19].

Problems

- 1 Show (1.1.20) by using the divergence theorem.
- 2 Show that $-\hat{r} \times (\hat{r} \times \vec{u}) = \hat{\theta}(\hat{\theta} \cdot \vec{u}) + \hat{\phi}(\hat{\phi} \cdot \vec{u}) = (\hat{\theta}\hat{\theta} + \hat{\phi}\hat{\phi}) \cdot \vec{u}$ where $\vec{u}$ is an arbitrary vector. We use the dyadic notation in the last expression.
- 3 For $R \neq 0$, show that $\vec{\nabla}(1/R) = -\vec{\nabla}'(1/R) = -\vec{R}/R^3$, where $\vec{R} = \vec{r} - \vec{r}'$ and $R = |\vec{R}|$. Here $\vec{\nabla}$ acts on $\vec{r}$, whereas $\vec{\nabla}'$ acts on $\vec{r}'$.
- 4 Show, in the sense of distributions, that $\vec{\nabla} \cdot (\vec{R}/R^3) = -\nabla^2(1/R) = 4\pi\delta(\vec{R})$. Verify separately that the ordinary divergence is zero for $R \neq 0$ and that the flux through any sphere enclosing $R = 0$ is 4π .
- 5 Let k and $\vec{u}$ be constants and let ψ satisfy the scalar Helmholtz equation (1.4.16). Show that $\vec{E} = \vec{\nabla} \times (\psi \vec{u})$ is divergence-free and satisfies the vector Helmholtz equation $(\nabla^2 + k^2)\vec{E} = 0$.
- 6 In a source-free, linear isotropic medium with differentiable, nonzero inhomogeneous permittivity $\epsilon(\vec{r})$ and constant permeability μ , define $k^2(\vec{r}) = \omega^2 \mu \epsilon(\vec{r})$. Show that $(\nabla^2 + k^2)\vec{E} = -\vec{\nabla}(\vec{E} \cdot \vec{\nabla} \ln \epsilon)$.

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